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XII - SET 1

COMMON QUARTERLY EXAMINATION – 2025

Standard XII

Reg.No. : | | | | | | |

MATHEMATICS

Time: 3.00 hrs. Part - I Marks: 90

I. Choose the correct answer: 20 x 1 = 20

1. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y$ is equal to

   a) $\frac{2\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π

2. If $\sin^{-1}x + \sin^{-1}y = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101}+y^{101}+z^{101}}$ is

   a) 0 b) 1 c) 2 d) 3

3. $\sin^{-1}(\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x) =$

   a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$

4. If $(AB)^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$, then $B^{-1} =$

   a) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$ b) $\begin{bmatrix} 8 & 5 \\ 3 & 2 \end{bmatrix}$ c) $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ d) $\begin{bmatrix} 8 & -5 \\ -3 & 2 \end{bmatrix}$

5. If A is a non-singular matrix such that $A^{-1} = \begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$ then $(A^T)^{-1} =$

   a) $\begin{bmatrix} -5 & 3 \\ 2 & 1 \end{bmatrix}$ b) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$ c) $\begin{bmatrix} -1 & -3 \\ 2 & 5 \end{bmatrix}$ d) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$

6. If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then the values of x and y are respectively

   a) $e^{(\Delta_2/\Delta_1)}, e^{(\Delta_3/\Delta_1)}$ b) $\log(\Delta_1/\Delta_3), \log(\Delta_2/\Delta_3)$

   c) $\log(\Delta_2/\Delta_1), \log(\Delta_3/\Delta_1)$ d) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$

7. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

   a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

8. If $x + y = k$ is normal to the parabola $y^2 = 12x$, Then the value of k is

   a) 3 b) –1 c) 1 d) 9

9. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

   a) 2ab b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

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10. The eccentricity of the ellipse $(x-3)^2+(y-4)^2=\frac{y^2}{9}$ is

a) $\frac{\sqrt{3}}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$

11. If $\left|Z-\frac{3}{2}\right|=2$, then the least value $|Z|$ is

a) 1 b) 2 c) 3 d) 5

12. If α,β and γ are the zeros of x^3+px^2+qx+r, then $\sum\frac{1}{\alpha}$ is

a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{r}$

13. The number of positive zeros of the polynomial $\sum_{r=0}^{n} nC_r(-1)^r x^r$ is

a) 0 b) n c) < n d) r

14. If $|z_1|=1, |z_2|=2, |z_3|=3$ and $|9z_1z_2+4z_1z_3+z_2z_3|=12$, then the value of $|z_1+z_2+z_3|$ is

a) 1 b) 2 c) 3 d) 4

15. If $(1+i)(1+2i)(1+3i)\ldots\ldots(1+ni)=x+iy$, then $2.5.10\ldots\ldots(1+n^2)$ is

a) 1 b) i c) x^2+y^2 d) $1+n^2$

16. If a vector $\vec{\alpha}$ lies in the plane of $\vec{\beta}$ and $\vec{\gamma}$ then

a) $\left[\vec{\alpha},\vec{\beta},\vec{\gamma}\right]=1$ b) $\left[\vec{\alpha},\vec{\beta},\vec{\gamma}\right]=-1$ c) $\left[\vec{\alpha},\vec{\beta},\vec{\gamma}\right]=0$ d) $\left[\vec{\alpha},\vec{\beta},\vec{\gamma}\right]=2$

17. The angle between the lines $\vec{r}=(\hat{i}+2\hat{j}-3\hat{k})+t(2\hat{i}+\hat{j}-2\hat{k})$ and the plane $\vec{r}.(\hat{i}+\hat{j})+4=0$ is

a) 0° b) 30° c) 45° d) 90°

18. Distance from the origin to the plane $3x-6y+2z+7=0$ is

a) 0 b) 1 c) 2 d) 3

19. If the direction cosines of a line are $\frac{1}{c},\frac{1}{c},\frac{1}{c}$, then

a) $c=\pm3$ b) $c=\pm\sqrt{3}$ c) $c>0$ d) $0<c<1$

20. If $\vec{a},\vec{b},\vec{c}$ are non-coplanar, non-zero vectors such that $\left[\vec{a},\vec{b},\vec{c}\right]=3$, then $\left[\vec{a}\times\vec{b},\vec{b}\times\vec{c},\vec{c}\times\vec{a}\right]^2$ is equal to

a) 81 b) 9 c) 27 d) 18

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. For what value of x the inequality $\frac{\pi}{2}<\cos^{-1}(3x-1)<\pi$ holds?

22. Find the value of $\sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right)$

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23. Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ by reducing it to row-echelon form.

24. Find the square root of $6 - 8i$

25. If $y = 2\sqrt{2x} + c$ is a tangent to the circle $x^2 + y^2 = 16$, find the value of c.

26. Find the equation of the ellipse with foci $(\pm 2, 0)$, vertices $(\pm 3, 0)$

27. Find the monic polynomial equation of minimum degree with real coefficients having $2 - \sqrt{3}i$ as a root.

28. A particle is acted upon by the forces $(3\hat{i} - 2\hat{j} + 2\hat{k})$ and $(2\hat{i} + \hat{j} - \hat{k})$ is displaced from the point (1,3,–1) to the point (4,–1,λ). If the work done by the force is 16 units, find the value of λ.

29. Find the angle between the plane $\vec{r}.(\hat{i} + \hat{j} - 2\hat{k}) = 3$ and $2x - 2y + z = 2$

30. Simplify the following : $i\, i^2\, i^3$ ..... i^{40}

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. Find the value of $\sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9} + \cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$

32. If $\text{adj}A = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.

33. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$

34. Find the length of Latus rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

35. Find the value of $\sum_{k=1}^{8}\left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9}\right)$

36. The parabolic communication antenna has a focus at 2m distance from the vertex of the antenna. Find the width of the antenna 3m from the vertex.

37. Solve the cubic equation : $2x^3 - x^2 - 18x + 9 = 0$ if sum of two of its roots vanishes.

38. Find the magnitude and direction cosinas of a moment of force $2\hat{i} + \hat{j} + \hat{k}$ about the point (2,0,–1) whose line of action passing through the origin.

39. Find the image of the point whose position vector is $\hat{i} + 2\hat{j} + 3\hat{k}$ in the plane $\vec{r}.(\hat{i} + 2\hat{j} + 4\hat{k}) = 38$

40. Verify $(AB)^{-1} = B^{-1}A^{-1}$ with $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}$

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Part - IV

IV. Answer all the questions. 7 x 5 = 35

41. a) Prove by vector method that $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

(OR)

b) Prove that the point of intersection of the tangents at 't_1' and 't_2' on the parabola $y^2 = 4ax$ is $[at_1t_2, a(t_1+t_2)]$.

42. a) If $z = x + iy$ is a complex numbers such that $\operatorname{Im}\left(\frac{2z+1}{iz+1}\right) = 0$, show that locus of z is $2x^2 + 2y^2 + x - 2y = 0$

(OR)

b) The upward speed v(t) of a rocket at a time t is approximated by $v(t) = at^2+bt+c$, $0 \le t \le 100$ where a,b and c are constant. It has been found that the speed at times t = 3, t = 6 and t = 9 seconds are respectively 64, 133 and 208 miles per second respectively. Find the speed at time t = 15 seconds. (Use Gaussian elimination method)

43. a) Determine the values of λ for which the following system of equations $x + y + 3z = 0$, $4x + 3y + \lambda z = 0$, $2x + y + 2z = 0$ has
(i) a unique solution (ii) a non-trivial solution

(OR)

b) Find the foci, vertices and length of major and minor axis of the conic $4x^2 + 36y^2 + 40x - 288y + 532 = 0$

44. a) Show that the lines $\vec{r} = (6\hat{i} + \hat{j} + 2\hat{k}) + s(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (3\hat{i} + 2\hat{j} - 2\hat{k}) + t(2\hat{i} + 4\hat{j} - 5\hat{k})$ are skew lines and hence find the shortest distance between them.

(OR)

b) Find the value of $\cot^{-1}(1) + \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - \sec^{-1}\left(-\sqrt{2}\right)$

45. a) Find the non-parametric form of Vector equation, and the Cartesian equations of the plane passing through the points (2,2,1), (9,3,6) and perpendicular to the plane $2x + 6y + 6z = 0$

(OR)

b) If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation $x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x + 140 = 0$, find all roots.

46. a) Assume the water issuing from the end of a horizontal pipe, 7.5m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2.5m below the line of the pipe, the flow of water has curved outward 3m beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?

(OR)

b) Solve the matrix inversion method :
$2x - y + 3z = 9$, $x + y + z = 9$, $x - y + z = 2$

47. a) Solve : $(x - 5)(x - 7)(x + 6)(x + 4) = 504$

(OR)

b) Solve the equation $z^3 + 8i = 0$ where $z \in C$

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